



# WTRTK-982 Product Specification

Product Specification: SPECIFICATION

Model: WTRTK-982

Description: Witte Intelligent WTRTK-982 high-precision positioning and orientation module

Enterprise quality system standard: ISO9001:2016 standard

Sensor production standard: GB/T191SJ 20873-2016

Product testing and testing standards: GB/T191SJ 20873-2016

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# 1 System introduction

## 1.1 Product Overview

The WTRTK-982 module is the general name of the 26X38MM size high-performance BDS/GNSS full-constellation high-precision positioning and orientation module series. This module product is based on the Nebulas IV design, a new generation of radio frequency baseband and high-precision algorithm integrated GNSS SoC chip independently developed by Hexinxingtong. The main antenna and slave antenna simultaneously track all system-wide frequency points including Beidou three global signals, achieving on-chip RTK positioning and dual-antenna directional solution. It has Dual-RTK dual RTK engine technology and differential input RTCM format adaptive recognition function. The built-in advanced anti-interference unit ensures that the module can still provide reliable and accurate positioning accuracy even in complex electromagnetic environments. Targeted at high-precision navigation and positioning fields such as drones, autonomous machines, precision agriculture and intelligent driving.





## 1.2 Product features

1. Small size, immersed gold technology, and excellent heat dissipation performance.
2. The antenna adopts  $\pi$ -type network impedance matching, the standing wave ratio is lower than 2.3, and the satellite search performance is strong.
3. Excellent positioning and directional function, high sensitivity, 1408 channels, supports full system full frequency point on-chip RTK positioning and dual-antenna directional calculation.
4. Dual interfaces: 1. TYPE-C interface, plug and play to directly connect to PC to view data. 2.XH2.54x6Pin interface, convenient for embedding into user system equipment.
5. Built-in antenna detection circuit.
6. Dual antenna input, Dual-RTK dual RTK engine technology, differential input RTCM format adaptive recognition.
7. Support 3 x UART, 1 x I2C, 1 x SPI, 1 x CAN.

## 1.3 Basic parameters

| parameter name name   | Parameter value                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| signal reception      | BDS/GPS/GLONASS/GALILEO/QZSS/SBAS                                                                       |
| Number of RF channels | Three-channel radio frequency supports simultaneous reception of all-constellation BDS, GPS and GLONASS |
| Cold start TTFF       | $\leq 30s$                                                                                              |
| Hot start TTFF        | $\leq 1s$                                                                                               |
| Recapture TTFF        | -30---85°C                                                                                              |

|                                  |                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------|
| Positioning accuracy             | Single point positioning: horizontal 1.5M, elevation 2.5M;                                   |
|                                  | Differential positioning: horizontal 0.4M+1ppm, elevation 0.8M+1ppm;                         |
|                                  | RTK positioning: horizontal 0.8cm+1ppm, elevation 1.5cm+1ppm.                                |
| Orientation accuracy             | 0.1 degree/1m baseline                                                                       |
| Time accuracy                    | 20ns                                                                                         |
| Speed measurement accuracy       | 0.03m/s                                                                                      |
| Data update rate                 | Positioning and direction finding 20Hz, 20Hz original observation                            |
| Serial port characteristics      | Baud rate range: 4800 bps ~921600 bps, default 115200bps, 8 data bits, no parity, 1 stop bit |
| Protocol                         | NMEA0183                                                                                     |
| Power supply                     | 5V                                                                                           |
| GPS&BD typical power consumption | 220mA &5V                                                                                    |
| Operating temperature            | -40 to +85 degrees Celsius                                                                   |
| Storage temperature              | -45 to +125 degrees Celsius                                                                  |
| Size                             | 26.0mm×38mm×7.6mm                                                                            |

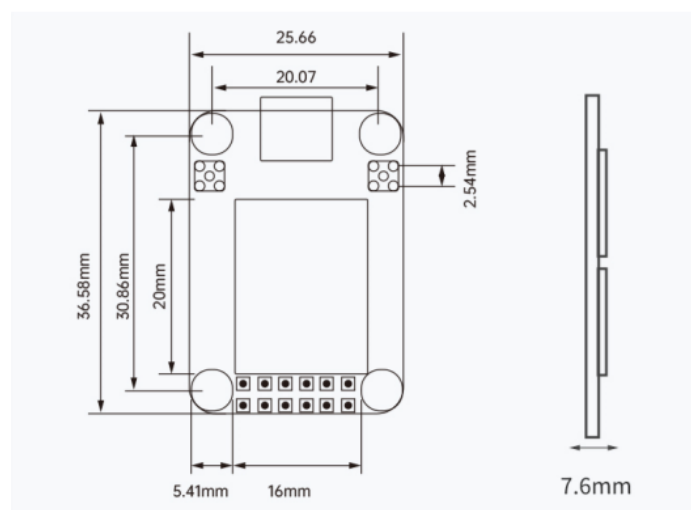
## 1.4 Product advantages

1. Excellent positioning and orientation function, supports BDS/GPS/GLONASS satellite system full frequency point on-chip RTK positioning and dual-antenna directional calculation, and supports QZSS and SBAS systems.
2. Positioning accuracy: RTK positioning: horizontal 0.8cm+1ppm, elevation 1.5cm+1ppm.
3. First positioning time:  $\leq 30s$ .
4. Built-in antenna detection and antenna short-circuit protection functions.

## 1.5 Product Application

1. Vehicle positioning and navigation;
2. Mobile phones, tablets, and handheld devices;
3. Embedded positioning equipment;
4. wearable equipment;
1. Drones, precision agriculture, robots.

## 1.6 Product size



## 2 How to use

### 2.1 Hardware connection

#### 2.1.1 USB connection

The module and computer can be directly connected using a USB data cable. The USB cable must be a data cable, that is, capable of transferring data and files. After connecting to the computer, you can check whether there is a serial port display in the device manager. If not, you need to install the CH340 driver. The following picture shows the CH340 driver installed. The device manager displays as follows:

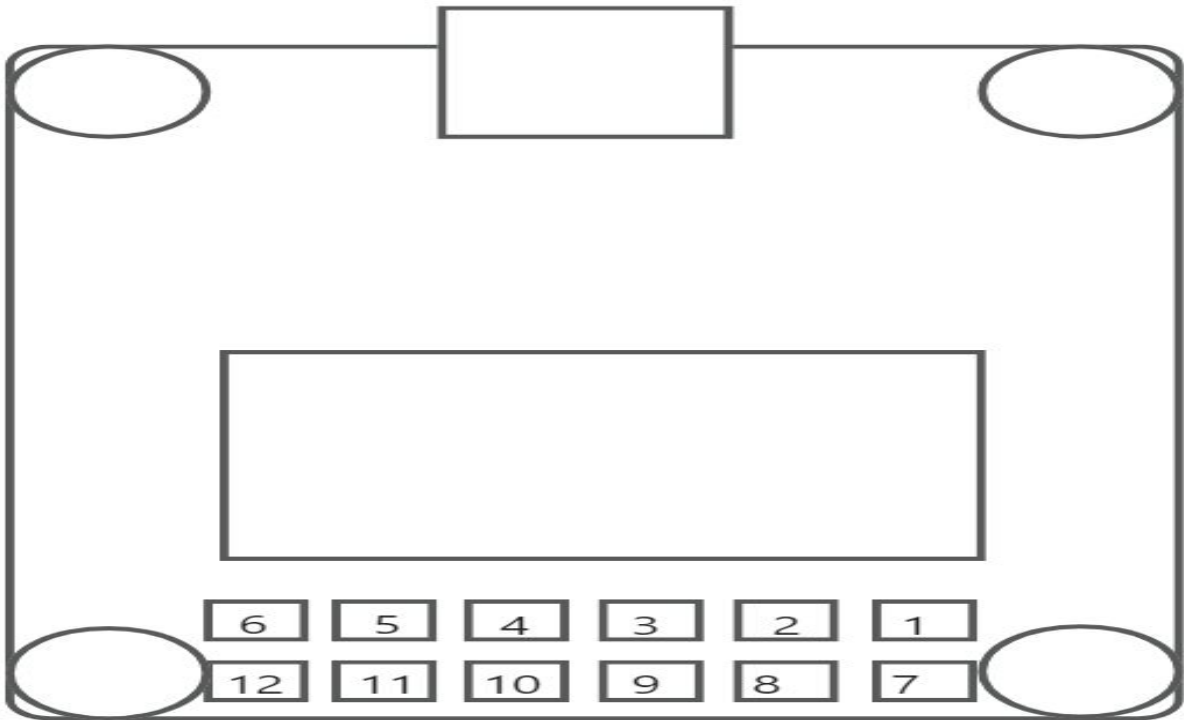


When installing the CP210X driver, the device manager displays as follows:

The three-in-one driver is CH340 and The six-in-one driver is CP2102 as follows:

[https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive\\_link](https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive_link)

## 2.1.2 Serial port connection





| Pin number | Pin name | Pin usage                                |
|------------|----------|------------------------------------------|
| 1          | PPS      | Second pulse                             |
| 2          | VCC      | The power supply must be connected to 5V |
| 3          | RXD      | Serial data input                        |
| 4          | TXD      | Serial data output                       |
| 5          | GND      | GND                                      |
| 6          | EN       | High level enable pin                    |
| 7          | PPS      | Second pulse                             |
| 8          | VCC      | The power supply must be connected to 5V |
| 9          | RX2      | Serial port 2 data input                 |
| 10         | TX2      | Serial port 2 data output                |
| 11         | GND      | GND                                      |
| 12         | EN       | High level enable pin                    |

Note: Pin 6 and pin 12 are enabled by high level by default and shut down when pulled low.

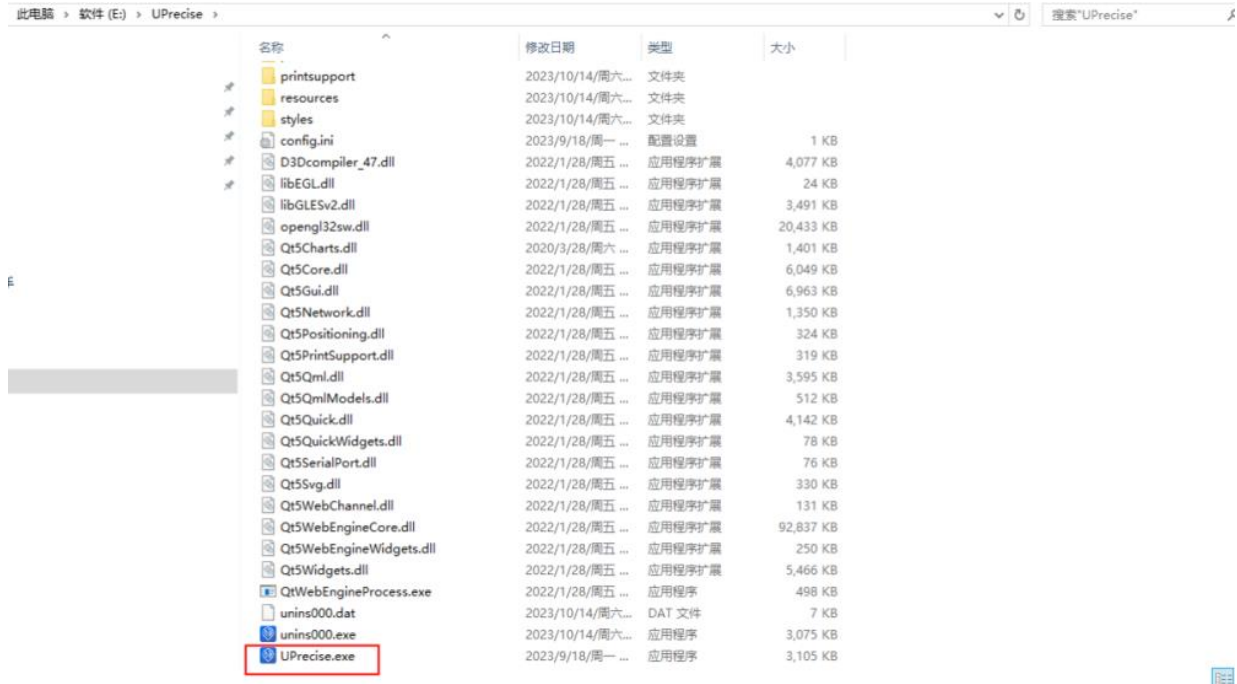
## 3 Software usage

### 3.1 Open the host computer

The PC software link is as follows:

[https://drive.google.com/file/d/1t3oI-u2BrwX-pxNWZKlIdX9PkmUlmeJA/view?usp=drive\\_link](https://drive.google.com/file/d/1t3oI-u2BrwX-pxNWZKlIdX9PkmUlmeJA/view?usp=drive_link)

After unzipping, open the folder and it will appear as follows:

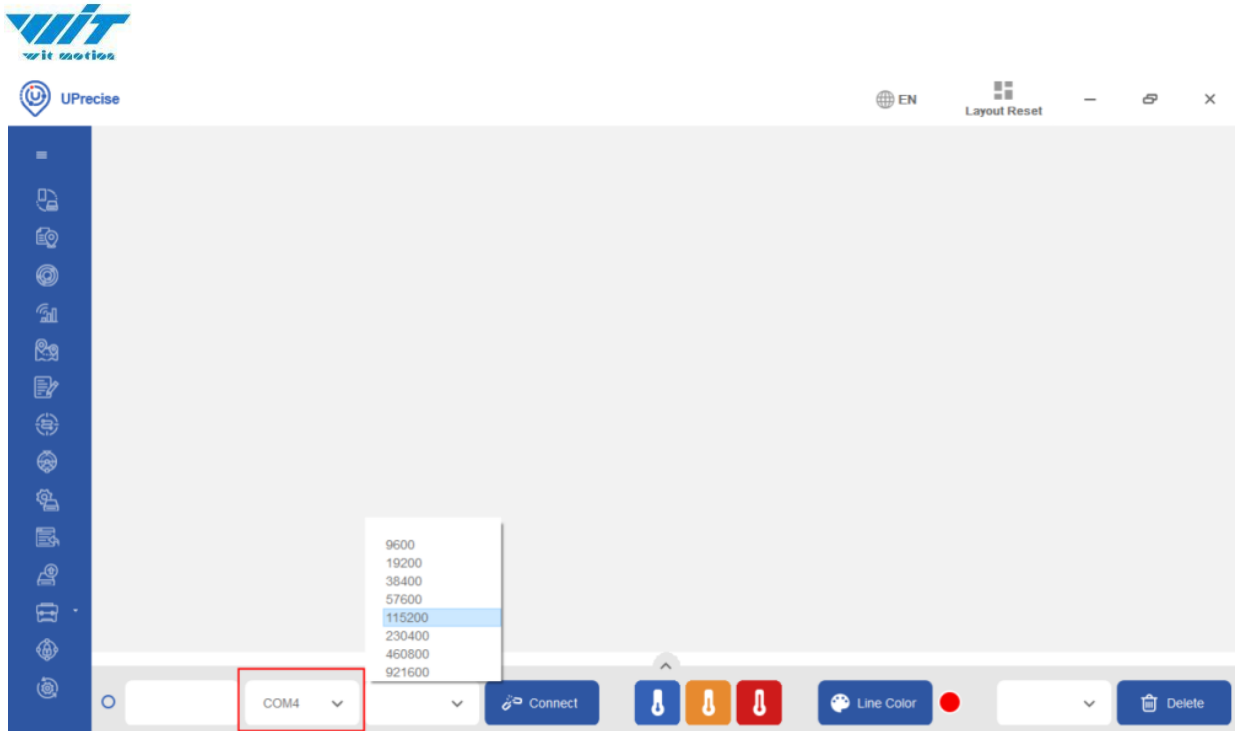


| 名称                      | 修改日期             | 类型     | 大小        |
|-------------------------|------------------|--------|-----------|
| printsupport            | 2023/10/14/周六... | 文件夹    |           |
| resources               | 2023/10/14/周六... | 文件夹    |           |
| styles                  | 2023/10/14/周六... | 文件夹    |           |
| config.ini              | 2023/9/18/周一...  | 配置设置   | 1 KB      |
| D3Dcompiler_47.dll      | 2022/1/28/周五...  | 应用程序扩展 | 4,077 KB  |
| libEGL.dll              | 2022/1/28/周五...  | 应用程序扩展 | 24 KB     |
| libGLESv2.dll           | 2022/1/28/周五...  | 应用程序扩展 | 3,491 KB  |
| opengl32sw.dll          | 2022/1/28/周五...  | 应用程序扩展 | 20,433 KB |
| Qt5Charts.dll           | 2020/3/28/周六...  | 应用程序扩展 | 1,401 KB  |
| Qt5Core.dll             | 2022/1/28/周五...  | 应用程序扩展 | 6,049 KB  |
| Qt5Gui.dll              | 2022/1/28/周五...  | 应用程序扩展 | 6,963 KB  |
| Qt5Network.dll          | 2022/1/28/周五...  | 应用程序扩展 | 1,350 KB  |
| Qt5Positioning.dll      | 2022/1/28/周五...  | 应用程序扩展 | 324 KB    |
| Qt5PrintSupport.dll     | 2022/1/28/周五...  | 应用程序扩展 | 319 KB    |
| Qt5Qml.dll              | 2022/1/28/周五...  | 应用程序扩展 | 3,595 KB  |
| Qt5QmlModels.dll        | 2022/1/28/周五...  | 应用程序扩展 | 512 KB    |
| Qt5Quick.dll            | 2022/1/28/周五...  | 应用程序扩展 | 4,142 KB  |
| Qt5QuickWidgets.dll     | 2022/1/28/周五...  | 应用程序扩展 | 78 KB     |
| Qt5SerialPort.dll       | 2022/1/28/周五...  | 应用程序扩展 | 76 KB     |
| Qt5Svg.dll              | 2022/1/28/周五...  | 应用程序扩展 | 330 KB    |
| Qt5WebChannel.dll       | 2022/1/28/周五...  | 应用程序扩展 | 131 KB    |
| Qt5WebEngineCore.dll    | 2022/1/28/周五...  | 应用程序扩展 | 92,837 KB |
| Qt5WebEngineWidgets.dll | 2022/1/28/周五...  | 应用程序扩展 | 250 KB    |
| Qt5Widgets.dll          | 2022/1/28/周五...  | 应用程序扩展 | 5,466 KB  |
| QtWebEngineProcess.exe  | 2022/1/28/周五...  | 应用程序   | 498 KB    |
| unins000.dat            | 2023/10/14/周六... | DAT 文件 | 7 KB      |
| unins000.exe            | 2023/10/14/周六... | 应用程序   | 3,075 KB  |
| UPrecise.exe            | 2023/9/18/周一...  | 应用程序   | 3,105 KB  |

Double-click UPrecise.exe to open the software.

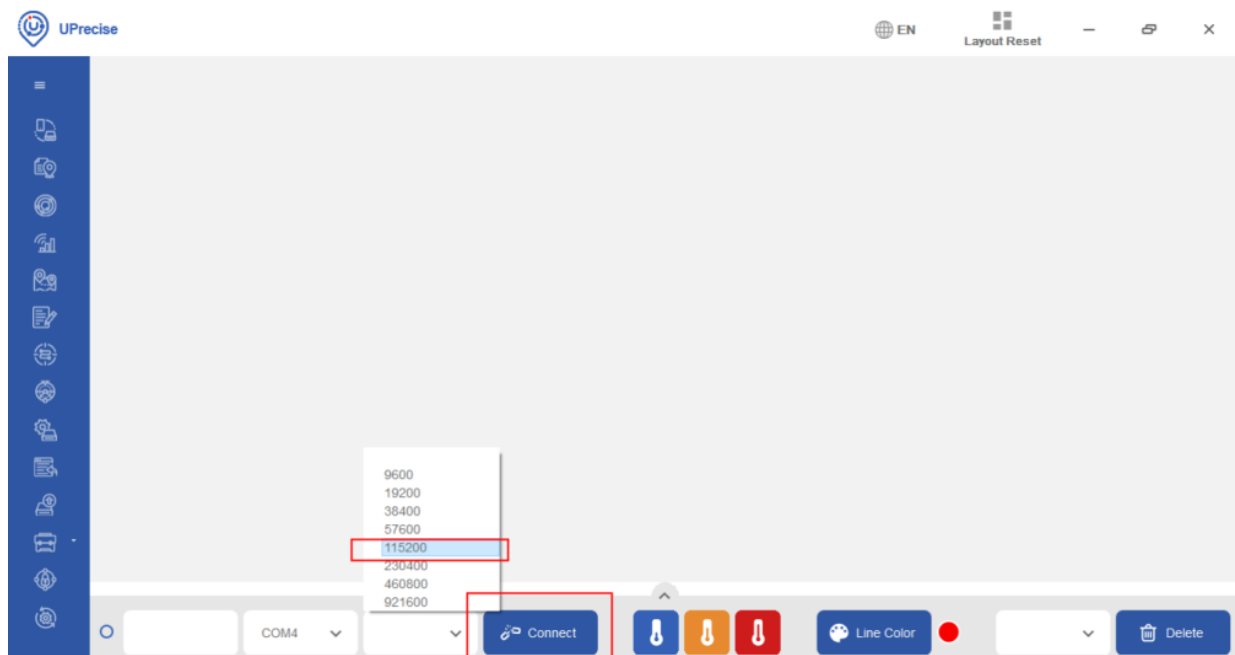
## 3.2 Select the corresponding port

The port number is the serial port displayed on the device manager interface. If you are not sure, you can plug and unplug the serial port module or USB data cable to see if the corresponding port displayed on the device manager port interface disappears.



### 3.3 Select baud rate

The default baud rate is 115200, click connect.



At this time, the host computer will not display any data. Find the input box below, check the full channel send and create a new row, and press the enter key to send the command.



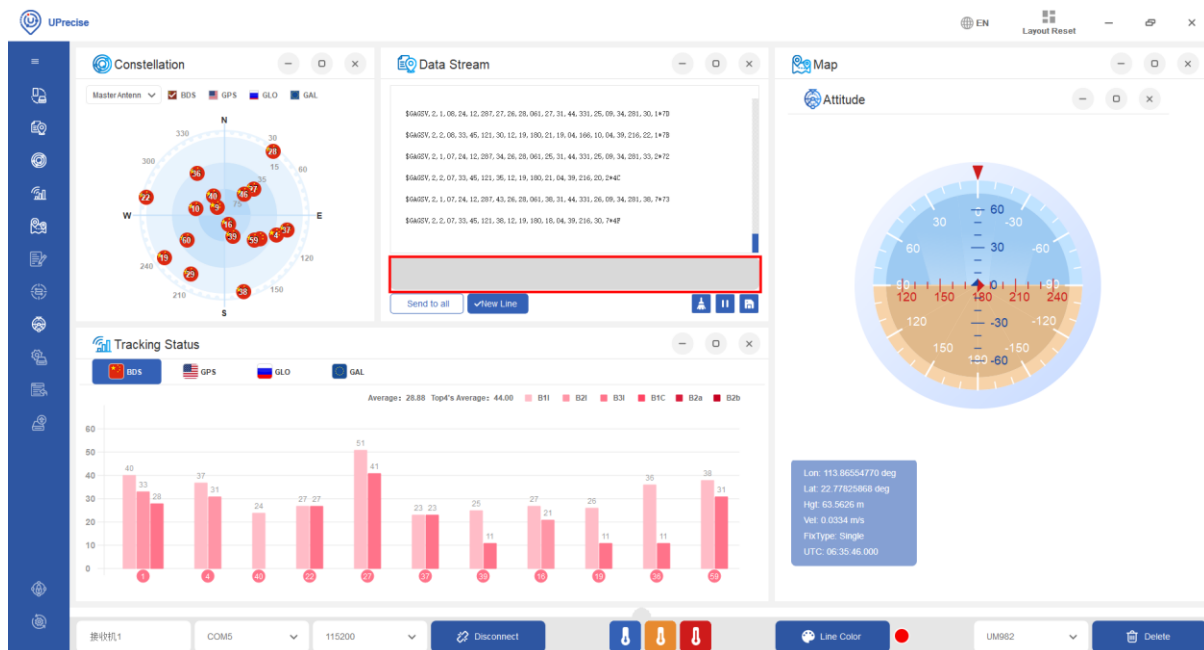
GNGGA 1

GNRMC 1

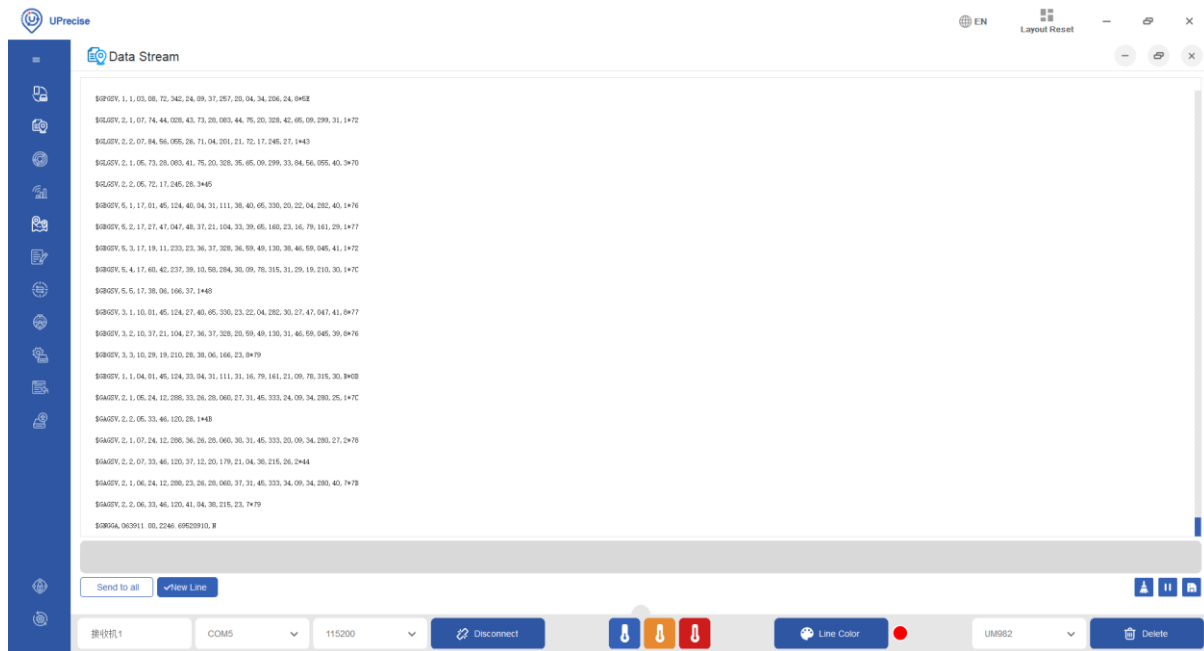
GNGSA 1

GNGSV 1

SAVECONFIG(Save), that's it. If you want to know more about the shortcut command operations and the meaning of the above statements, please read the protocol specification.

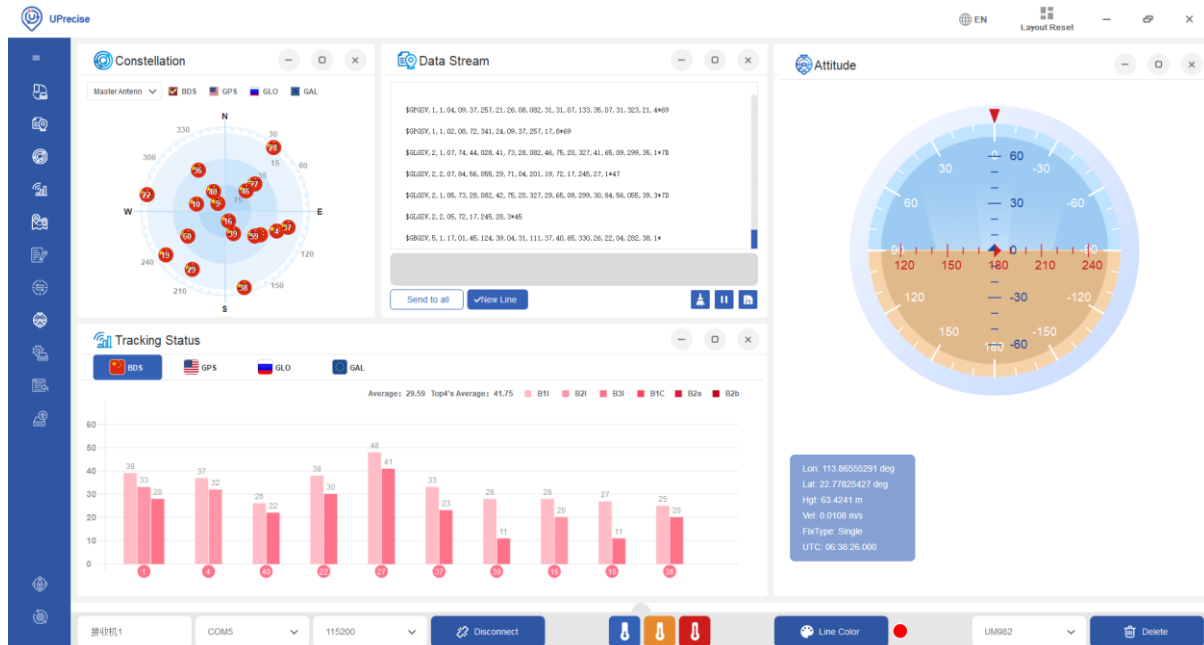


## 3.4 View raw data



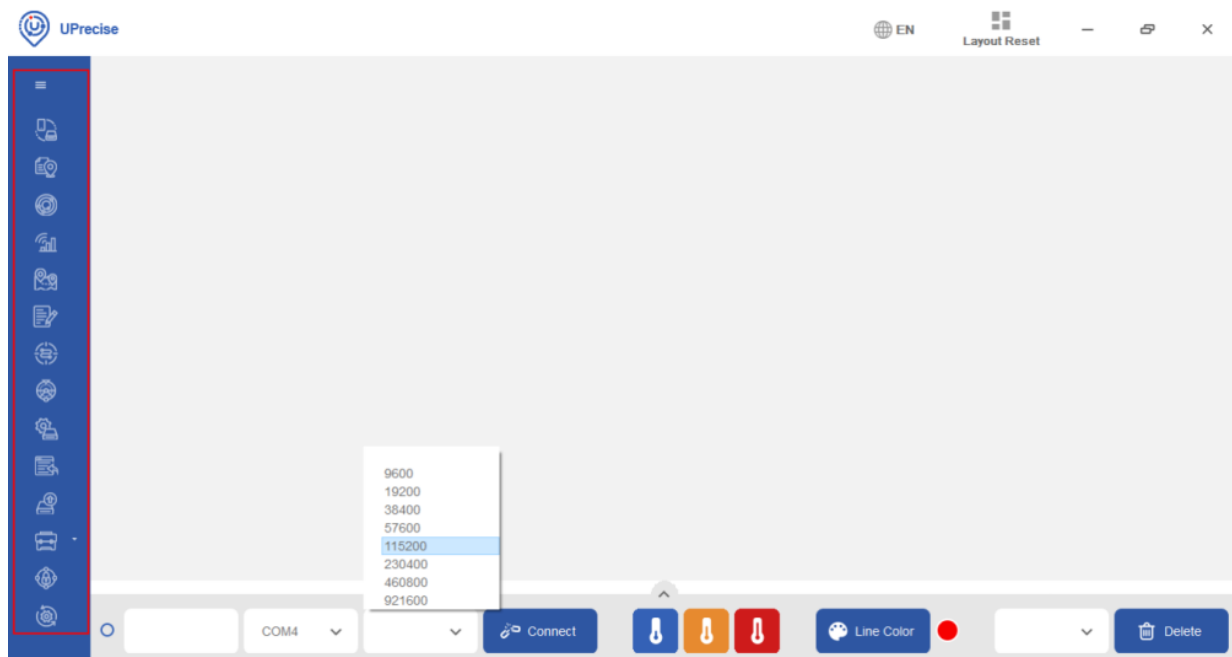
## 3.5 View all data

The original data is not intuitive enough. Click View on the left menu bar to select the data type you want to view.



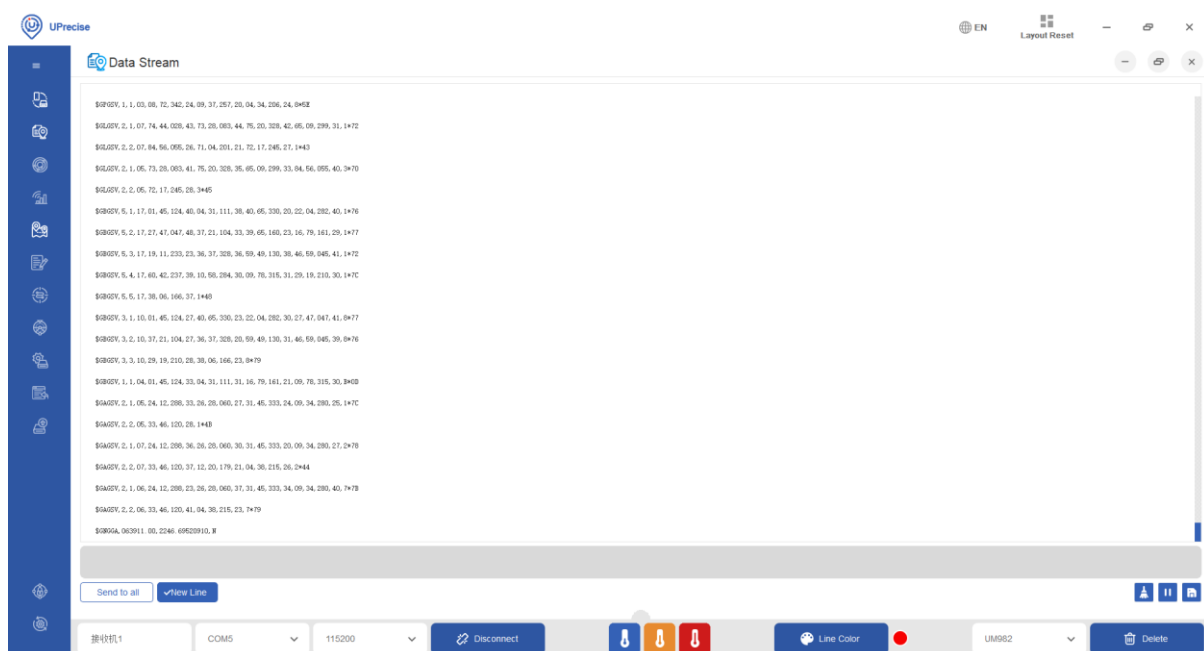
## 3.6 Main interface functions

Displayed as follows:

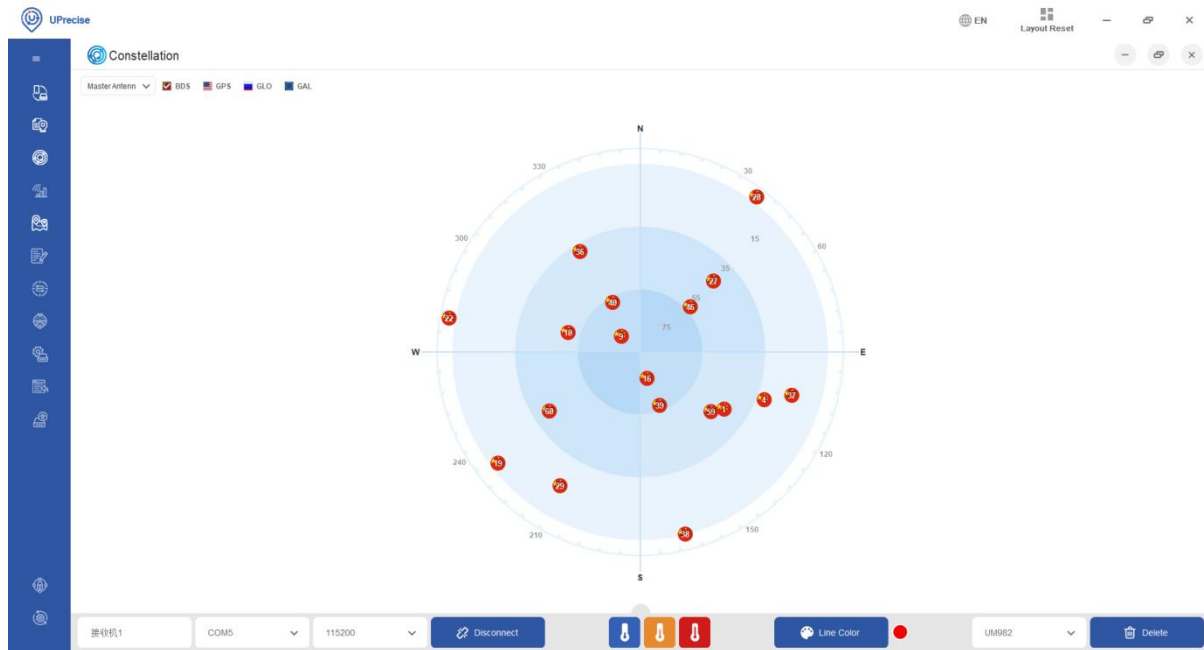


## 3.7 View data display

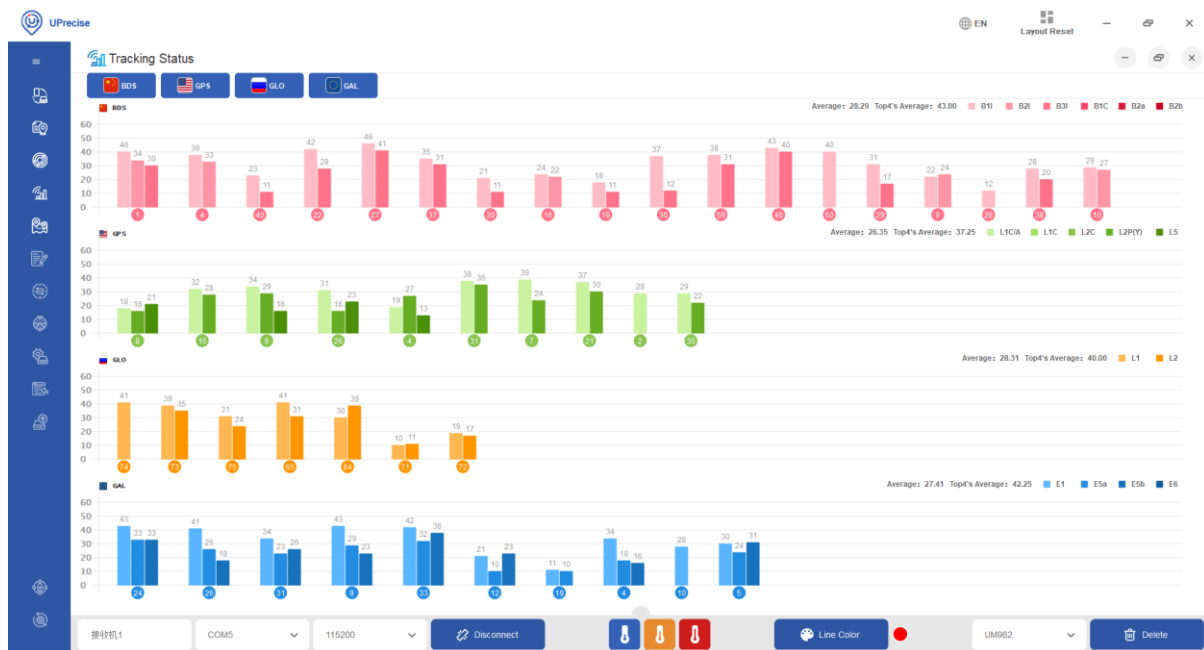
### 1. Original data interaction



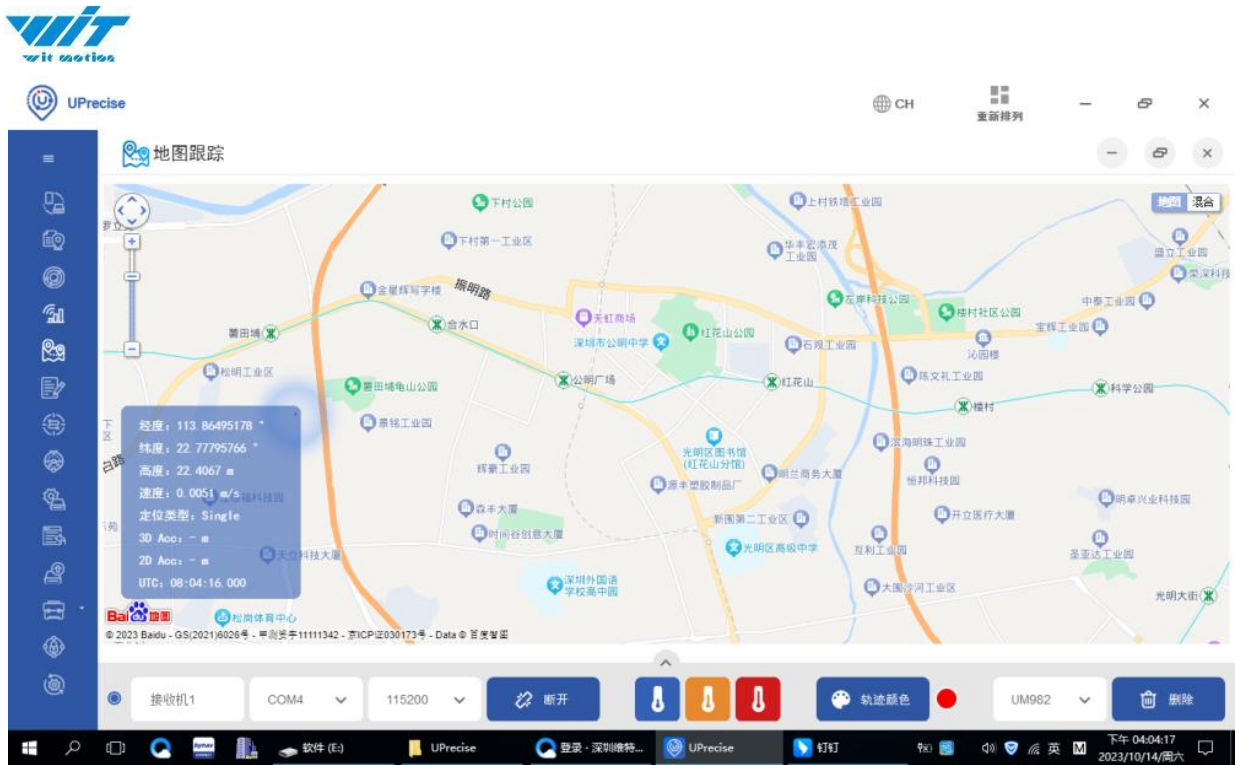
## 2. Satellite distribution view



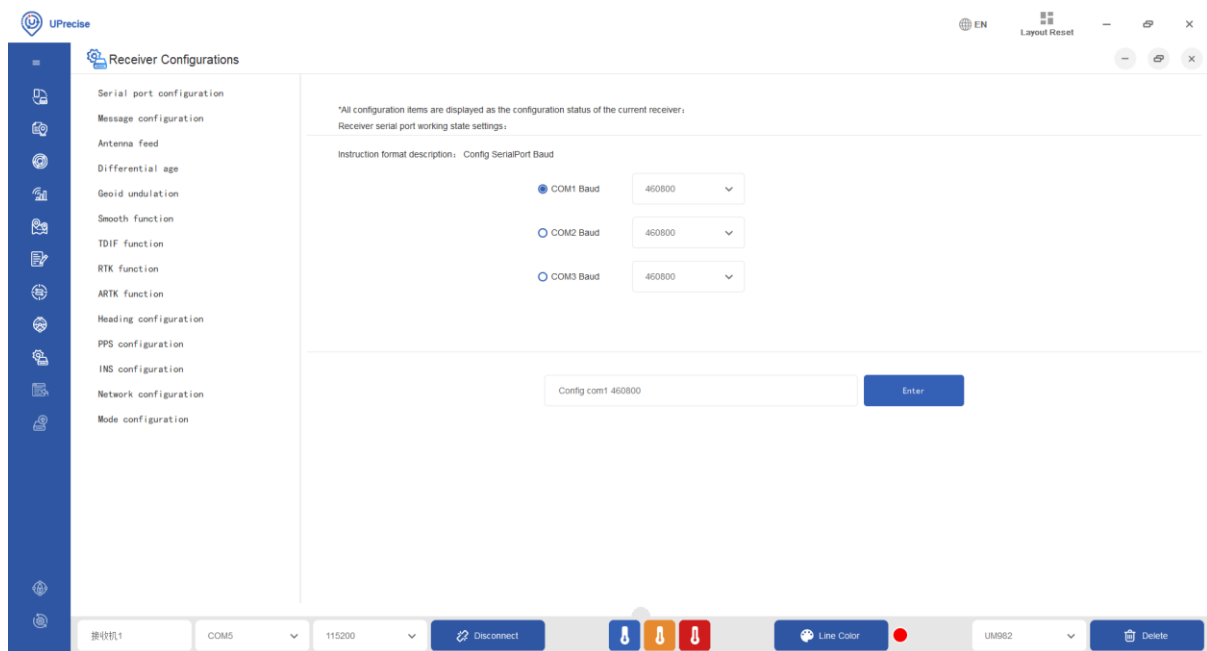
## 3. Satellite tracking: satellite signal strength



## 4. Map tracking: Display the surrounding map and module location, and output the current latitude and longitude, altitude, speed, positioning type, and UTC time.



## 5. Receiver settings: Take changing the baud rate as an example



## 6. Statement analysis



**WIT motion**

UPrecise

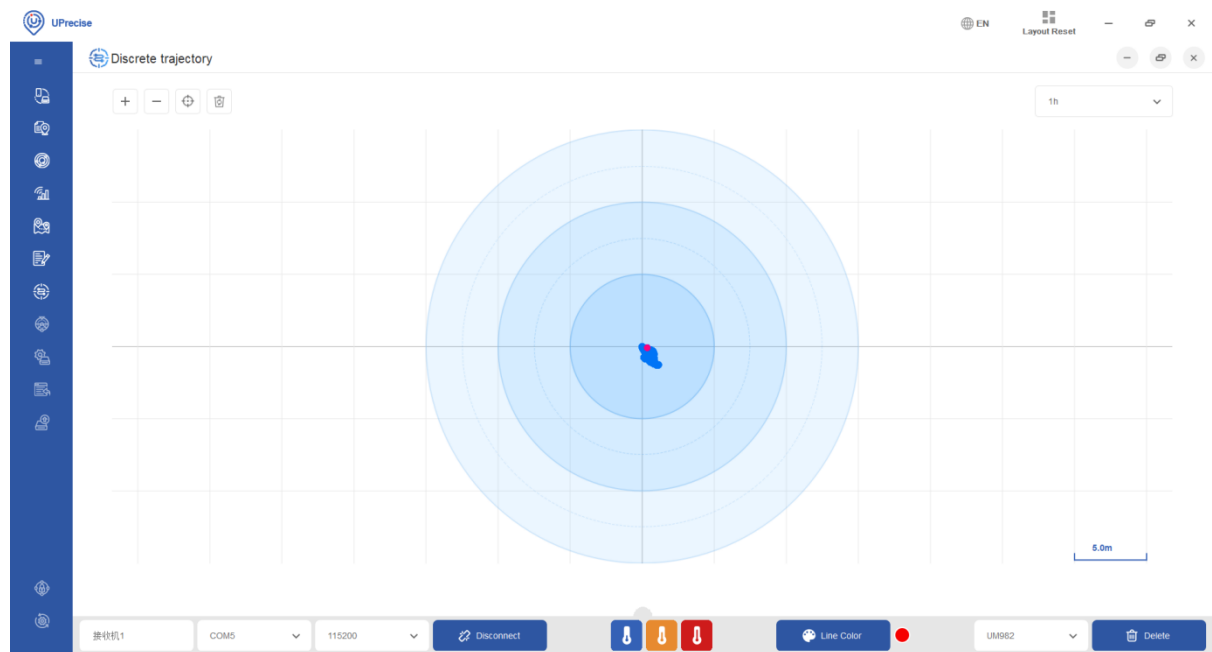
Message

Parameter Value Unit Description

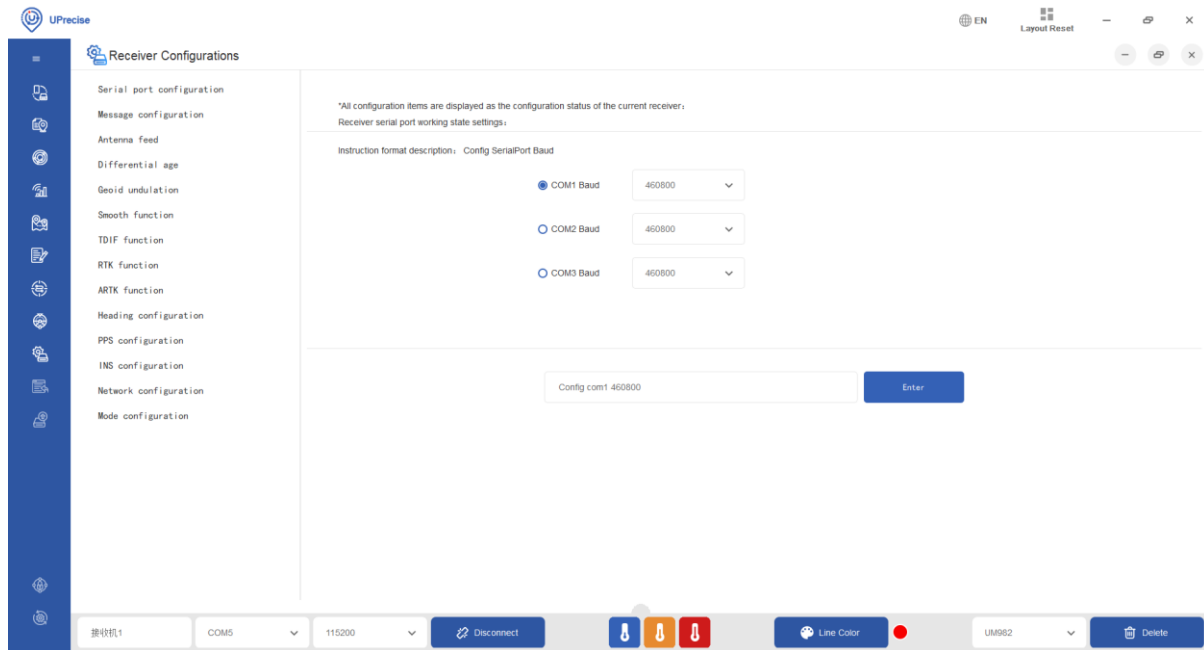
|            |                |             |                                                                                                                                                         |
|------------|----------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| utc        | 064816.00      | hh:mm:ss    | Coordinated Universal Time (UTC)                                                                                                                        |
| lat        | 2246.69578873  | ddmm.mmmmm  | Latitude                                                                                                                                                |
| lat dir    | N              |             | N-North latitude, S-South latitude                                                                                                                      |
| lon        | 11351.93297283 | ddddd.mmmmm | Longitude                                                                                                                                               |
| lon dir    | E              |             | E-East longitude, W-West longitude                                                                                                                      |
| qual       | 1              |             | 0-Invalid 1-Fix valid 2-Differential mode, fix valid 3-PPS mode, fix valid 4-RTK mode 5-Float RTK 6-Estimated mode 7-Manual input mode 8-Simulator mode |
| # sats     | 22             |             | Number of satellites in use                                                                                                                             |
| hdop       | 0.7            |             | Horizontal Dilution of Precision                                                                                                                        |
| Alt        | 66.6429        |             | Altitude re:mean-sea-level                                                                                                                              |
| e-units    | M              |             | m Meters                                                                                                                                                |
| undulation | -4.1001        |             | Geoidal separation =ellipsoid height - height above mean-sea-level                                                                                      |
| u-units    | M              |             | m Meters                                                                                                                                                |
| age        |                |             |                                                                                                                                                         |
| stn ID     |                |             |                                                                                                                                                         |

接收机1 COM5 115200 Disconnect Line Color UM982 Delete

## 7. Discrete trajectory tracking



## 8. Posture information



## 3.8 RTK solution

### 3.8.1

Clamp the test clip on the pin header interface of WTRTK-982 or solder the pin header and wire it out. The type-c end supplies power, the TX, RX, GND of the ttl serial port, and the TX, RX, GND of the module. TX-RX, RX-TX, GND to GND, make sure the wiring is correct.

### 3.8.2

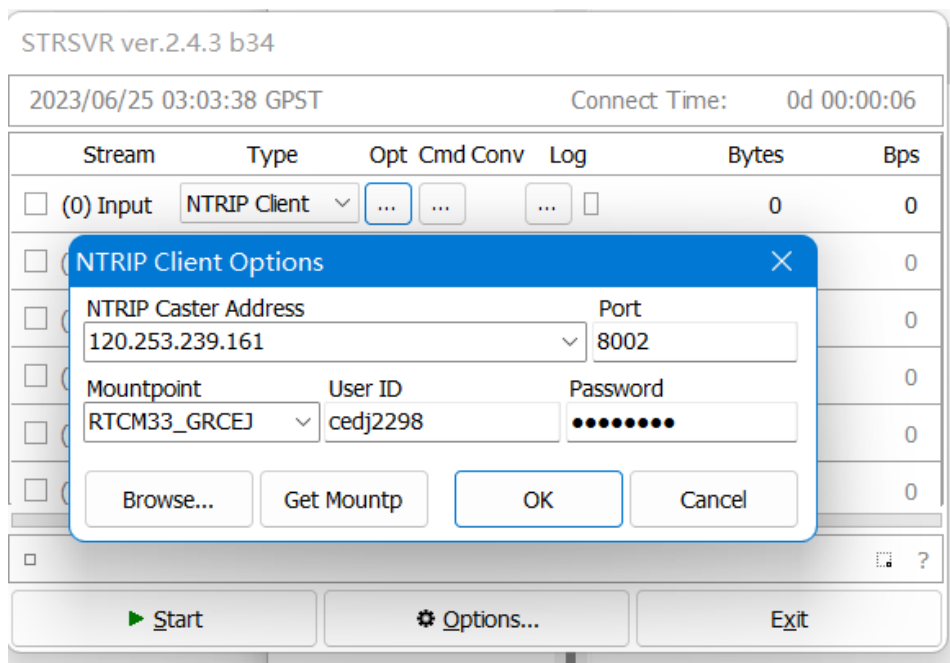
The application link is below:

[https://drive.google.com/file/d/1uGk2MxBG27oq3fe\\_yfdJ0Qg2-PBTx-4/view?usp=drive\\_link](https://drive.google.com/file/d/1uGk2MxBG27oq3fe_yfdJ0Qg2-PBTx-4/view?usp=drive_link)

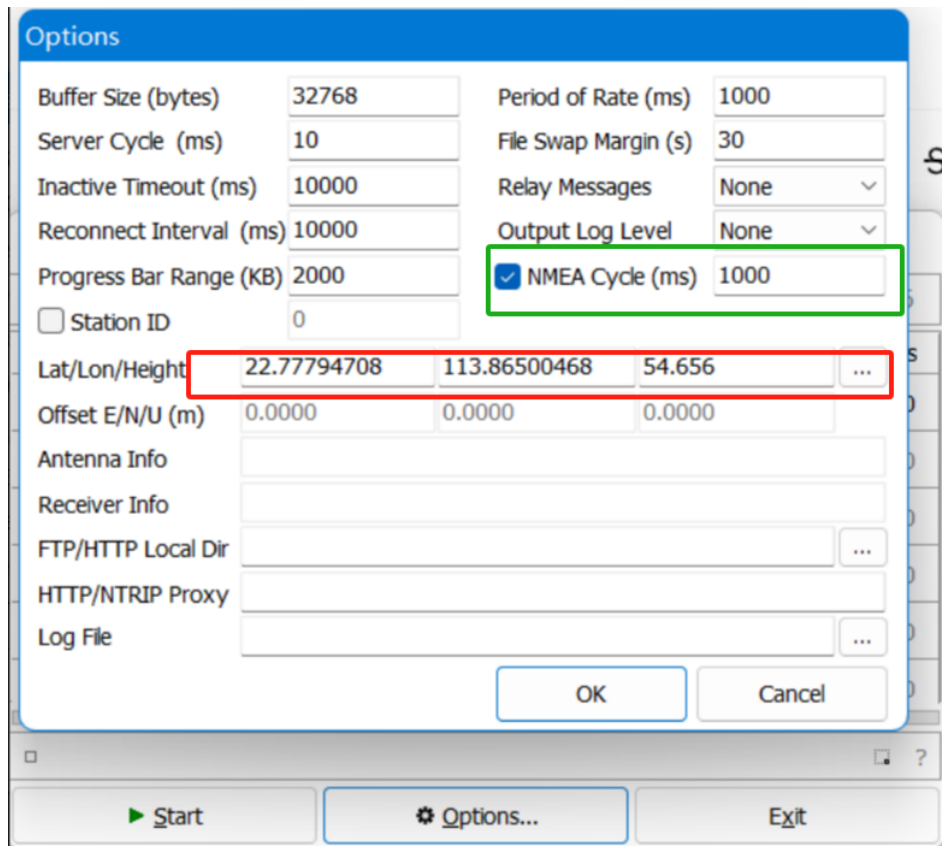
Click strsvr to open the strsvr application;

| 名称                                                                   | 修改日期            | 类型                 | 大小        |
|----------------------------------------------------------------------|-----------------|--------------------|-----------|
| N4_Upgrade工具svn2194                                                  | 2023/6/20 17:39 | 文件夹                |           |
| strsvr                                                               | 2023/6/21 14:05 | 文件夹                |           |
| uSTAR_R3.0.0_win32_Build2100                                         | 2023/6/25 10:56 | 文件夹                |           |
| 1095408-UM982eb-V1R0-20210831_pcb.brd                                | 2022/5/9 19:54  | BRD 文件             | 4,168 KB  |
| N4_Upgrade工具svn2194.zip                                              | 2023/6/20 17:32 | 360压缩 ZIP 文件       | 1,253 KB  |
| strsvr.zip                                                           | 2023/6/21 14:04 | 360压缩 ZIP 文件       | 2,773 KB  |
| UM982eb_User Manual_CH_P1.0.1.pdf                                    | 2022/5/9 19:54  | Microsoft Edge ... | 2,767 KB  |
| um982eb-v1r0-20220422.pdf                                            | 2022/5/9 19:54  | Microsoft Edge ... | 370 KB    |
| UM982的参考设计.rar                                                       | 2023/6/19 14:23 | 360压缩 RAR 文件       | 3,860 KB  |
| UM982规格参数.pdf                                                        | 2023/6/19 14:22 | Microsoft Edge ... | 1,397 KB  |
| Unicore Reference Commands Manual For N4 High Precision Products_... | 2023/6/19 14:22 | Microsoft Edge ... | 2,629 KB  |
| UPrecise V1.0.639.exe                                                | 2023/6/19 14:11 | 应用程序               | 55,725 KB |
| uSTAR_R3.0.0_win32_Build2100.rar                                     | 2023/6/20 10:09 | 360压缩 RAR 文件       | 42,657 KB |

Click Opt, move, or use Qianxun account. The following uses mobile as an example.  
After entering, click OK



Click Options and enter the longitude, latitude and altitude of the current location at the red box position. For the specific longitude and latitude, you can go to the host computer to view the gga data package. The longitude and latitude are accurate to eight decimal places; the green box position is to modify the return rate, the maximum return rate. It is 20hz, which corresponds to 50ms.

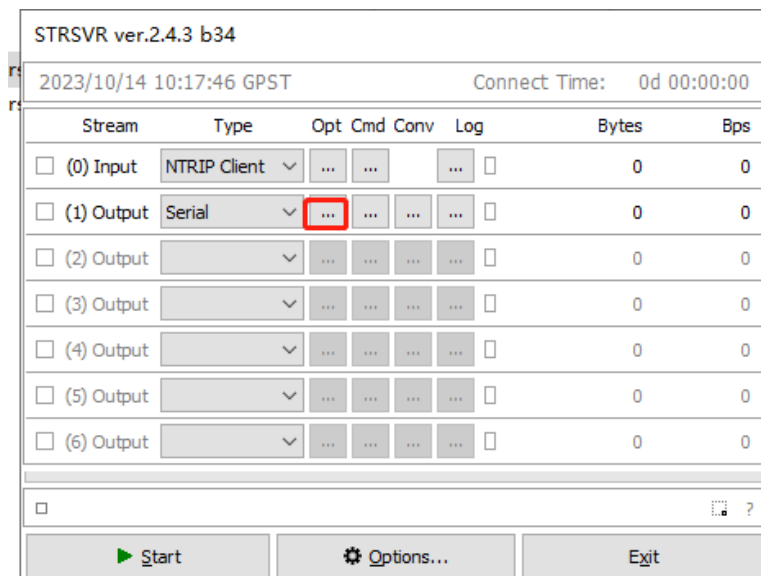


**Options**

|                                     |             |                                                     |        |
|-------------------------------------|-------------|-----------------------------------------------------|--------|
| Buffer Size (bytes)                 | 32768       | Period of Rate (ms)                                 | 1000   |
| Server Cycle (ms)                   | 10          | File Swap Margin (s)                                | 30     |
| Inactive Timeout (ms)               | 10000       | Relay Messages                                      | None   |
| Reconnect Interval (ms)             | 10000       | Output Log Level                                    | None   |
| Progress Bar Range (KB)             | 2000        | <input checked="" type="checkbox"/> NMEA Cycle (ms) | 1000   |
| <input type="checkbox"/> Station ID | 0           |                                                     |        |
| Lat/Lon/Height                      | 22.77794708 | 113.86500468                                        | 54.656 |
| Offset E/N/U (m)                    | 0.0000      | 0.0000                                              | 0.0000 |
| Antenna Info                        |             |                                                     |        |
| Receiver Info                       |             |                                                     |        |
| FTP/HTTP Local Dir                  | ...         |                                                     |        |
| HTTP/NTRIP Proxy                    |             |                                                     |        |
| Log File                            | ...         |                                                     |        |

OK Cancel

For Output configuration, select Serial for Type.



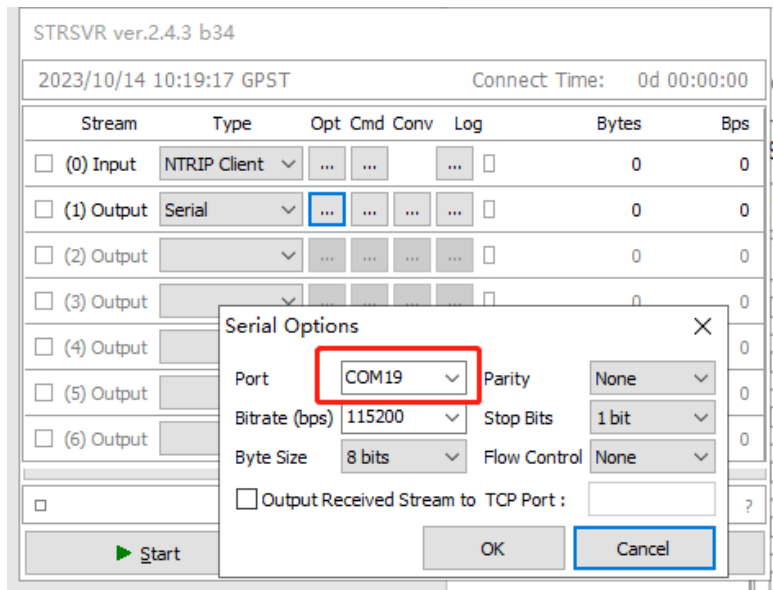
STRSVR ver.2.4.3 b34

2023/10/14 10:17:46 GPST Connect Time: 0d 00:00:00

| Stream                              | Type         | Opt | Cmd | Conv | Log                      | Bytes | Bps |
|-------------------------------------|--------------|-----|-----|------|--------------------------|-------|-----|
| <input type="checkbox"/> (0) Input  | NTRIP Client | ... | ... | ...  | <input type="checkbox"/> | 0     | 0   |
| <input type="checkbox"/> (1) Output | Serial       | ... | ... | ...  | <input type="checkbox"/> | 0     | 0   |
| <input type="checkbox"/> (2) Output | ...          | ... | ... | ...  | <input type="checkbox"/> | 0     | 0   |
| <input type="checkbox"/> (3) Output | ...          | ... | ... | ...  | <input type="checkbox"/> | 0     | 0   |
| <input type="checkbox"/> (4) Output | ...          | ... | ... | ...  | <input type="checkbox"/> | 0     | 0   |
| <input type="checkbox"/> (5) Output | ...          | ... | ... | ...  | <input type="checkbox"/> | 0     | 0   |
| <input type="checkbox"/> (6) Output | ...          | ... | ... | ...  | <input type="checkbox"/> | 0     | 0   |

Start Options... Exit

Click on the blue box, go in and see the port option box. For this com port, select the com port corresponding to the ttl end. You can check it in the device manager for details.



Click Start and see that the progress bar turns green. At this time, the TX on the three-in-one side will flash.

### 3.8.3

At this time, you can see the GNGGA statement output of the original data of the host computer.

消息输出。  
\$GNGGAH,054536.00,4004.73794643,N,11614.19884494,E,4,39,0.5,65.7579,M,-  
8.4923,M,02,0\*2C

Just change the number next to E to 4. At this time, you can also find that the RTK light is always on.

**Note:** Since domestic and overseas accounts are different, 3.8.2-3.8.3 are for reference only.

## 4. Supporting documents

UM982 product manual



[https://drive.google.com/file/d/1a1S-j7aOD4XACeDrdLNMDIOeu10k665Z/view?usp=drive\\_link](https://drive.google.com/file/d/1a1S-j7aOD4XACeDrdLNMDIOeu10k665Z/view?usp=drive_link)

NMEA protocol

[https://drive.google.com/file/d/1BIVXX3PZGr\\_NAcUOdflzvWp\\_O8L\\_fEzc/view?usp=drive\\_link](https://drive.google.com/file/d/1BIVXX3PZGr_NAcUOdflzvWp_O8L_fEzc/view?usp=drive_link)